AQ3.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

If $(1, 2, 3) + v = (3, 2, 1)$(1, 2, 3) + v = (3, 2, 1) in the vector space R^3 R3 with usual addition and scalar multiplication, where $v = (a, b, c)$v = (a, b, c), then find the value of $a + b + c$a + b + c.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0

1 point

Which of the following statements are related to the cancellation law for vectors x, y, zx, y, z in a vector space.

Statement P:Statement P: $x + y = z + y \implies x = z$.x + y = z + y $\implies$ x = z.
Statement Q:Statement Q: $x + y = z + y \implies x = y$.x + y = z + y $\implies$ x = y.
Statement R:Statement R: $x + y = z + y \implies y = z$.x + y = z + y $\implies$ y = z.
Statement S:Statement S: $x + y + z = x \implies y + z = 0$x + y + z = x $\implies$ y + z = 0

Find the number of correct statements

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2

1 point

1 point

Which of the following options is/are true for a non zero vector space VV ?

☐ $x + y = z + y \implies x = z \; \forall x, y, z \in V$x + y = z + y $\implies$ x = z ∀x, y, z ∈ V .

☐ $ax = bx, \; \forall \; x \in V \implies a = b,$ where $a, b \in R$ax = bx, ∀ x ∈ V $\implies$ a = b, where a, b ∈ R.

☐ $ax = ay, \; \forall \; a \in R \implies x = y,$ where $x, y \in V$.ax = ay, ∀ a ∈ R $\implies$ x = y, where x, y ∈ V .

☐ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$x + y = z + y \implies x = z \; \forall x, y, z \in V$x + y = z + y $\implies$ x = z ∀x, y, z ∈ V .
$ax = bx, \; \forall \; x \in V \implies a = b,$ where $a, b \in R$ax = bx, ∀ x ∈ V $\implies$ a = b, where a, b ∈ R.
$ax = ay, \; \forall \; a \in R \implies x = y,$ where $x, y \in V$.ax = ay, ∀ a ∈ R $\implies$ x = y, where x, y ∈ V .

1 point

Which of the following options is/are true for a non zero vector space VV ?

☐ $av = 0, \; \forall \; v \in V \implies a = 0$av = 0, ∀ v ∈ V $\implies$ a = 0.

☐

$av = 0, \forall\, a \in R \implies v = 0$av = 0, ∀ a ∈ R ⟹ v = 0.

☐ $c \in R$c ∈ R and $v, w \in V$v, w ∈ V such that, $c(v - w) \neq cv - cw$c(v − w) = cv − cw.

☐ $c((v + w) - (v - w) - 2w) = 0\ \forall c \in R,$ and $v, w \in V$c((v + w) − (v − w) − 2w) = 0 ∀c ∈ R, and v, w ∈ V

No, the answer is incorrect.
Score: 0
Accepted Answers:
$av = 0, \forall\, v \in V \implies a = 0$av = 0, ∀ v ∈ V ⟹ a = 0.
$av = 0, \forall\, a \in R \implies v = 0$av = 0, ∀ a ∈ R ⟹ v = 0.
$c((v + w) - (v - w) - 2w) = 0\ \forall c \in R,$ and $v, w \in V$c((v + w) − (v − w) − 2w) = 0 ∀c ∈ R, and v, w ∈ V

Let $(3, 2, -3) \in R^3$(3, 2, −3) ∈ R3 be a vector from the vector space R^3 R3 with usual addition and scalar multiplication. If $c(3, 2, -3) = (-6, -4, 6)$c(3, 2, −3) = (−6, −4, 6), find the value of cc.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -2
1 point
1 point
Which of the following options is/are true?
☐ The zero vector is a unique vector in a vector space.
☐ Inverse of a vector is a unique vector in a vector space.
☐ $0v \neq 00$v = 0, where vv is a vector from a vector space of RR.
☐ $c0 = 0$c0 = 0, where $c \in R$c ∈ R and 0 is the zero vector from a vector space over RR.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The zero vector is a unique vector in a vector space.
Inverse of a vector is a unique vector in a vector space.
$c0 = 0$c0 = 0, where $c \in R$c ∈ R and 0 is the zero vector from a vector space over RR.

Level 2

Let VV be a plane plane parallel to the XYXY -plane. Any plane parallel to XYXY -plane is given by $z = c$z = c. We define addition of two vectors $v_1 = (x_1, y_1, c)$v1 = (x1, y1, c) and $v_2 = (x_2, y_2, c)$ v2 = (x2, y2, c) on VV as follows: First project v_1 v1 and v_2 v2 on the XYXY -plane (we will get the vectors $(x_1, y_1, 0)$(x1, y1, 0) and $(x_2, y_2, 0)$(x2, y2, 0) by projection on the XYXY -plane) and then calculate the addition of the vectors we obtained by the projection on the XYXY -plane (we will obtain $(x_1 + x_2, y_1 + y_2, 0)$(x1 + x2, y1 + y2, 0)). Then project the obtained vector back to the plane VV (we will obtain the vector $(x_1 + x_2, y_1 + y_2, c)$(x1 + x2, y1 + y2, c)).
Let VV be a plane which is parallel to the XYXY -plane, defined by $z = 2$z = 2. Let $v_1 = (1, 2, 2)$v1

= (1, 2, 2) and v_2 = (0, 3, 2)v2 = (0, 3, 2) be in VV . Answer questions 7 and 8.

If $v_1^{'}$ = (a_1, b_1, c_1)v1′ = (a1, b1, c1) and $v_2^{'}$ = (a_2, b_2, c_2)v2′ = (a2, b2, c2) are the projections of v_1 v1 and v_2 v2 on the XYXY - plane, respectively, then find the value of
2(a_1 + b_1 + c_1) − (a_2 + b_2 + c_2)2(a1 + b1 + c1) − (a2 + b2 + c2).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

If v_1 + v_2 = (a_1, b_1, c_1)v1 + v2 = (a1, b1, c1) as per the addition defined above, then find the value of (2a_1 + b_1 − 2c_1)(2a1 + b1 − 2c1).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

Let VV be a plane plane parallel to the XYXY -plane. Any plane parallel to the XYXY -plane is given by z = cz = c. We define scalar multiplication of a vector v = (x, y, c)v = (x, y, c) on VV as follows: First project vv on the XYXY -plane first (we will get the vector (x, y, 0)(x, y, 0) by projection on the XYXY -plane) and then calculate the scalar (α)(α) multiple of the vector we obtained by the projection on the XYXY -plane (we will obtain (αx, αy, 0)(αx, αy, 0)). Then project the obtained vector back to the plane VV (we will obtain the vector (αx, αy, c)(αx, αy, c)).
Let VV be a plane which is parallel to the XYXY -plane, defined by z = 2z = 2. Let v = (1, 2, 2)v = (1, 2, 2) be in VV . Answer questions 9 and 10.

If $v^{'}$ = (a_1, b_1, c_1)v′ = (a1, b1, c1) is the projection of vv on the XYXY - plane, then find the value of (a_1 + 2b_1 + c_1)(a1 + 2b1 + c1).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 5
1 point

If 4v = (a_1, b_1, c_1)4v = (a1, b1, c1) as per the scalar multiplication defined above, then find the value of (a_1 + 2b_1 + c_1)(a1 + 2b1 + c1).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 22
1 point

Check Answers
Your score is: 0/10